

2.4 Sensitivity of Optimisation Algorithms to Initialisation

2 Gradient Descent

Question 1.

At stationary points of f_θ , we have

$$f'_\theta(x) = 4x^3 - 3x - \cos(\theta) = 0$$

Observe that $x \in [-1, 1]$, so we can write $x = \cos(\phi)$, where $\phi \in [0, \pi]$. Since $\cos(3\phi) = 4\cos^3(\phi) - 3\cos(\phi)$, we obtain

$$\cos(3\phi) - \cos(\theta) = 0$$

We have $\theta \in (0, \pi)$ and $3\phi \in [0, 3\pi]$. We therefore have

$$3\phi = \theta \quad \text{or} \quad 3\phi = 2\pi - \theta \quad \text{or} \quad 3\phi = 2\pi + \theta$$

The stationary points are then given by

$$x_1^* = \cos\left(\frac{2\pi + \theta}{3}\right), \quad x_2^* = \cos\left(\frac{2\pi - \theta}{3}\right), \quad x_3^* = \cos\left(\frac{\theta}{3}\right)$$

where $x_1^* < x_2^* < x_3^*$.

We look at the second derivative to determine the nature of these stationary points. We have

$$f''_\theta(x) = 12x^2 - 3$$

and so there is a minimum at x_i^* if $|x_i^*| > \frac{1}{2}$ and a maximum at x_i^* if $|x_i^*| < \frac{1}{2}$.

For $\theta \in (0, \pi)$, we have:

1. $\frac{2\pi + \theta}{3} \in (\frac{2\pi}{3}, \pi)$, so $\cos\left(\frac{2\pi + \theta}{3}\right) \in (-1, -\frac{1}{2})$ and there is a minimum at $x_1^* = \cos\left(\frac{2\pi + \theta}{3}\right) < 0$.
2. $\frac{2\pi - \theta}{3} \in (\frac{\pi}{3}, \frac{2\pi}{3})$, so $\cos\left(\frac{2\pi - \theta}{3}\right) \in (-\frac{1}{2}, \frac{1}{2})$ and there is a maximum at $x_2^* = \cos\left(\frac{2\pi - \theta}{3}\right)$.
3. $\frac{\theta}{3} \in (0, \frac{\pi}{3})$, so $\cos\left(\frac{\theta}{3}\right) \in (\frac{1}{2}, 1)$ and there is a minimum at $x_3^* = \cos\left(\frac{\theta}{3}\right) > 0$.

Plotting the graph of $f_\theta(x)$ for $\theta = \frac{\pi}{6}, \frac{3\pi}{4}$, we see that there are two minima and a maximum in the interval $x \in [-1, 1]$, as expected.

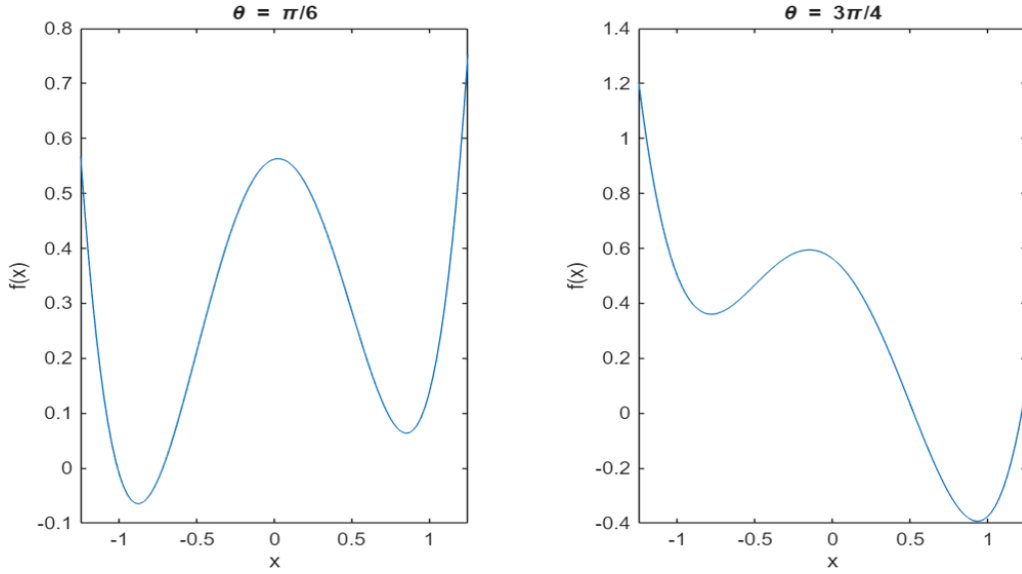


Figure 1: $f_\theta(x)$ against x for $\theta = \frac{\pi}{6}, \frac{3\pi}{4}$ and $x \in [-\frac{5}{4}, \frac{5}{4}]$

Here, we have plotted $f_\theta(x)$ for the range $x \in [-\frac{5}{4}, \frac{5}{4}]$ to make the stationary points easier to see. To verify that all stationary points shown in Figure (1) lie in $[-1, 1]$, we note that $4x^3 - 3x = \cos(\theta) \in (-1, 1)$ only if $x \in [-1, 1]$, so all stationary points shown do indeed lie in the given domain.

Question 2.

For $\theta = \frac{\pi}{6}$, the two minima are located at

$$x_1^* = \cos\left(\frac{13\pi}{18}\right) = -0.642787609687...$$

$$x_3^* = \cos\left(\frac{\pi}{18}\right) = 0.984807753012...$$

The maximum is located at

$$x_2^* = \cos\left(\frac{11\pi}{18}\right) = -0.342020143326...$$

In one dimension, we have

$$\nabla f_\theta(x) = f'_\theta(x) = 4x^3 - 3x - \frac{\sqrt{3}}{2}$$

x_0	x_{1000}	$\min\{ x_{1000} - x_1 , x_{1000} - x_3 \}$
-1	-0.64278761	3.6207×10^{-10}
-0.8	-0.6427876099	2.4482×10^{-10}
-0.7	-0.6427876098	1.1967×10^{-10}
-0.68	-0.6427876098	8.3414×10^{-11}
-0.66	-0.6427876097	4.1527×10^{-11}
-0.64	-0.6427876097	7.2744×10^{-12}
-0.62	-0.6427876096	6.4679×10^{-11}
-0.6	-0.6427876096	1.3295×10^{-10}
-0.4	-0.642787605	4.7246×10^{-9}
-0.38	-0.642787601	8.6454×10^{-9}
-0.36	-0.6427875861	2.3593×10^{-8}
-0.34	0.984807753	5.5511×10^{-16}
-0.32	0.984807753	5.5511×10^{-16}
-0.3	0.984807753	5.5511×10^{-16}
-0.2	0.984807753	5.5511×10^{-16}
0	0.984807753	5.5511×10^{-16}
0.2	0.984807753	5.5511×10^{-16}
0.4	0.984807753	5.5511×10^{-16}
0.6	0.984807753	5.5511×10^{-16}
0.8	0.984807753	5.5511×10^{-16}
0.9	0.984807753	5.5511×10^{-16}
0.92	0.984807753	5.5511×10^{-16}
0.94	0.984807753	5.5511×10^{-16}
0.96	0.984807753	5.5511×10^{-16}
0.98	0.984807753	5.5511×10^{-16}
1	0.984807753	6.6613×10^{-16}

Table 1: Outcomes of gradient descent for $\theta = \frac{\pi}{6}$, $N = 1000$, $x_0 \in \left\{\frac{k}{50}\right\}_{k=-50}^{50}$

From Table (1), we observe that the outcome x_{1000} depends on the initial value x_0 . For $x_0 < x_2^*$, we get $x_{1000} \approx x_1^*$, and for $x_0 > x_2^*$, we get $x_{1000} \approx x_3^*$. This is as expected: in the region $[-1, x_2^*) \ni x_1^*$, the derivative f'_θ always “points away” from the minimum at x_1^* (that is, $f'_\theta(x) < 0$ for $x \in [-1, x_1^*)$ and $f'_\theta(x) > 0$ for $x \in (x_1^*, x_2^*)$), and so $x_t = x_{t-1} - hf'_\theta(x_{t-1})$ approaches x_1^* . Similarly, in the region $(x_2^*, 1] \ni x_3^*$, the derivative always “points away” from the minimum at x_3^* , and so the sequence x_t approaches x_3^* instead.

We can formalise this intuition as follows. Let $x^* \in [a, b]$ be a minimum of f_θ , and suppose we have $\text{sgn}(f'_\theta(x)) = \text{sgn}(x - x^*)$ for all $x \in [a, b] \setminus \{x^*\}$. Observe that, for $x \in [a, b]$

$$\begin{aligned}
f'_\theta(x) &= 4x^3 - 3x - \cos(\theta) \\
&= 4(x^3 - [x^*]^3) - 3(x - x^*) + f'_\theta(x^*) \\
&= (x - x^*)[4(x^2 + xx^* + [x^*]^2) - 3] \quad \text{since } f'_\theta(x^*) = 0
\end{aligned}$$

We define $g(x; x^*) = 4(x^2 + xx^* + [x^*]^2) - 3$. Now $\text{sgn}(f'_\theta(x)) = \text{sgn}(x - x^*) \implies g(x; x^*) > 0$. Also, since $x, x^* \in [a, b] \subseteq [-1, 1]$, we must have $g(x; x^*) \leq 9$. Then, for $x_{t-1} \in [a, b]$, we have

$$\begin{aligned}
|x_t - x^*| &= |x_{t-1} - hf'_\theta(x_{t-1}) - x^*| \\
&= |x_{t-1} - x^*||1 - hg(x; x^*)| \\
&< |x_{t-1} - x^*| \quad \text{for } 0 < h < \frac{2}{9}
\end{aligned}$$

Further suppose that we have $m \leq g(x; x^*) \leq M$ for $x \in [a, b]$ and $0 < m < M \leq 9$. For $h = 0.01 < \frac{1}{M}$, we then have $1 - Mh \leq 1 - hg(x; x^*) \leq 1 - mh$, and so $\text{sgn}(x_t - x^*) = \text{sgn}(x_{t-1} - x^*)$. Then x_t lies between

x_{t-1} and x^* , which lie in $[a, b]$. Thus

$$\begin{aligned} |x_t - x^*| &\in [(1 - Mh)|x_{t-1} - x^*|, (1 - mh)|x_{t-1} - x^*|] \\ &\subseteq [(1 - Mh)^t|x_0 - x^*|, (1 - mh)^t|x_0 - x^*|] \text{ by induction} \\ \implies |x_t - x^*| &\rightarrow 0 \text{ as } t \rightarrow \infty \end{aligned} \tag{I}$$

and x_t converges to x^* for $x_0 \in [a, b]$. Additionally, $\text{sgn}(x_t - x^*) = \text{sgn}(x_0 - x^*)$ for all $t \in \mathbb{N}$.

Note that $g(x; x^*) = \frac{f'_\theta(x)}{x - x^*} \forall x \neq x^*$ when x^* is a stationary point of f'_θ ; in particular, $g(x_2^*; x^*) = 0$. Now observe that $\frac{dg}{dx} = 4(2x + x^*)$. We refer to our numerical values for x_1^*, x_2^*, x_3^* :

1. For $x^* = x_1^*$ and $x \leq x_2^*$, $\frac{dg}{dx} \leq 4(2x_2^* + x_1^*) < 0$, so $g(x; x_1^*)$ is strictly decreasing on $[-1, x_2^*]$ with a root at $x = x_2^*$. Then, if $x_0 < x_2^*$, we can take $[a, b] = [-1, \max\{x_1^*, x_0\}]$ and $m = \min\{g(x_1^*; x_1^*), g(x_0; x_1^*)\} > 0$, $M = 9$ to see that x_t converges to x_1^* .
2. For $x^* = x_3^*$ and $x \geq x_2^*$, $\frac{dg}{dx} \geq 4(2x_2^* + x_3^*) > 0$, so $g(x; x_3^*)$ is strictly increasing on $[x_2^*, 1]$ with a root at $x = x_2^*$. Then, if $x_0 > x_2^*$, we can take $[a, b] = [\min\{x_3^*, x_0\}, 1]$ and $m = \min\{g(x_3^*; x_3^*), g(x_0; x_3^*)\} > 0$, $M = 9$ to see that x_t converges to x_3^* .

Thus we can see that, for any starting point $x_0 \in [-1, 1] \setminus x_2^*$, x_t converges to a minimum of f_θ , explaining our observation above.

Also from Table (1), we observe that for $x_0 \in [-1, x_2^*)$, the distance of the final value x_{1000} from the minimum x_1^* appears to depend on x_0 , while for $x_0 \in (x_2^*, x_3^*)$, the distance of x_{1000} from x_3^* appears to be constant (we exclude the region $x_0 \in (x_3^*, 1]$ in this analysis, since we only have one data point). We hypothesise that the rate of convergence is higher in the interval (x_2^*, x_3^*) , so the error has stabilised after 1000 steps (and the nonzero error observed is likely from truncation errors). Meanwhile, in the interval $[-1, x_2^*)$, the rate of convergence is lower, and so the error in the 1000th term has not yet stabilised, so it decreases noticeably as the starting value x_0 gets closer to the minimum at x_1^* .

3 The Monte Carlo Method

Question 3.

We have

$$\begin{aligned} \nu &= \mathbf{E}[g(X)] \\ \hat{\nu}_N &\triangleq \frac{1}{N} \sum_{i=1}^N g(X^i) \end{aligned}$$

where the samples $\{X^i\}_{i=1}^N$ are independent and identically distributed (iid) and distributed like X .

Then

$$\begin{aligned} \mathbf{E}[\hat{\nu}_N] &= \frac{1}{N} \sum_{i=1}^N \mathbf{E}[g(X^i)] \\ &= \frac{1}{N} \sum_{i=1}^N \mathbf{E}[g(X)] \text{ since } X^i \text{ is distributed like } X \\ &= \mathbf{E}[g(X)] = \nu \end{aligned}$$

so $\hat{\nu}_N$ is unbiased for ν , as required.

Suppose $\mathbf{Var}[g(X)] < \infty$. Then

$$\begin{aligned}\mathbf{Var}[\hat{\nu}_N] &= \frac{1}{N^2} \sum_{i=1}^N \mathbf{Var}[g(X^i)] \quad \text{since the } X^i \text{ are independent} \\ &= \frac{1}{N^2} \sum_{i=1}^N \mathbf{Var}[g(X)] \quad \text{since } X^i \text{ is distributed like } X\end{aligned}$$

and so

$$\mathbf{Var}[\hat{\nu}_N] = \frac{1}{N} \mathbf{Var}[g(X)] \quad (\text{II})$$

Question 4.

Define the sample mean $\hat{\mu}_N^h$ and sample variance $(\hat{\sigma}^2)_N^h$ to be

$$\begin{aligned}\hat{\mu}_N^h &\triangleq \frac{1}{N} \sum_{i=1}^N X_{Th^{-1}}^{h,i} \\ (\hat{\sigma}^2)_N^h &\triangleq \frac{1}{N} \sum_{i=1}^N \left[\left(X_{Th^{-1}}^{h,i} \right)^2 - \left(\hat{\mu}_N^h \right)^2 \right]\end{aligned}$$

where the samples $\left\{ X_{Th^{-1}}^{h,i} \right\}_{i=1}^N$ are iid and distributed like $X_{Th^{-1}}^h$. By Question (3), $\hat{\mu}_N^h$ is an unbiased estimator of $\mu = \mathbf{E}[X_{Th^{-1}}^h]$. Also,

$$\begin{aligned}\mathbf{E}[(\hat{\sigma}^2)_N^h] &= \frac{1}{N} \sum_{i=1}^N \mathbf{E} \left[\left(X_{Th^{-1}}^{h,i} \right)^2 - \left(\hat{\mu}_N^h \right)^2 \right] \\ &= \mathbf{E} \left[\left(X_{Th^{-1}}^h \right)^2 - \mu^2 + \mu^2 - \left(\hat{\mu}_N^h \right)^2 \right] \\ &= \mathbf{Var}[X_{Th^{-1}}^h] - \mathbf{Var}[\hat{\mu}_N^h] \\ &= \left(1 - \frac{1}{N} \right) \mathbf{Var}[X_{Th^{-1}}^h] \quad \text{using Equation (II)}\end{aligned}$$

Thus $(\hat{\sigma}^2)_N^h$ is an asymptotically unbiased estimator of $\sigma^2 = \mathbf{Var}[X_{Th^{-1}}^h]$ as $N \rightarrow \infty$.

k	h	$\hat{\mu}_N^h$	$(\hat{\sigma}^2)_N^h$
0	0.1	0.34591	0.65289
1	0.05	0.34567	0.65299
2	0.025	0.34627	0.65273
3	0.0125	0.34529	0.65315
4	0.00625	0.34599	0.65285
5	0.003125	0.35074	0.65077
6	0.0015625	0.34456	0.65347
7	0.00078125	0.34283	0.65421
8	0.000390625	0.35896	0.64707
9	0.0001953125	0.33609	0.65704
10	0.00009765625	0.33364	0.65805

Table 2: $\hat{\mu}_N^h$ and $(\hat{\sigma}^2)_N^h$ for each value of k

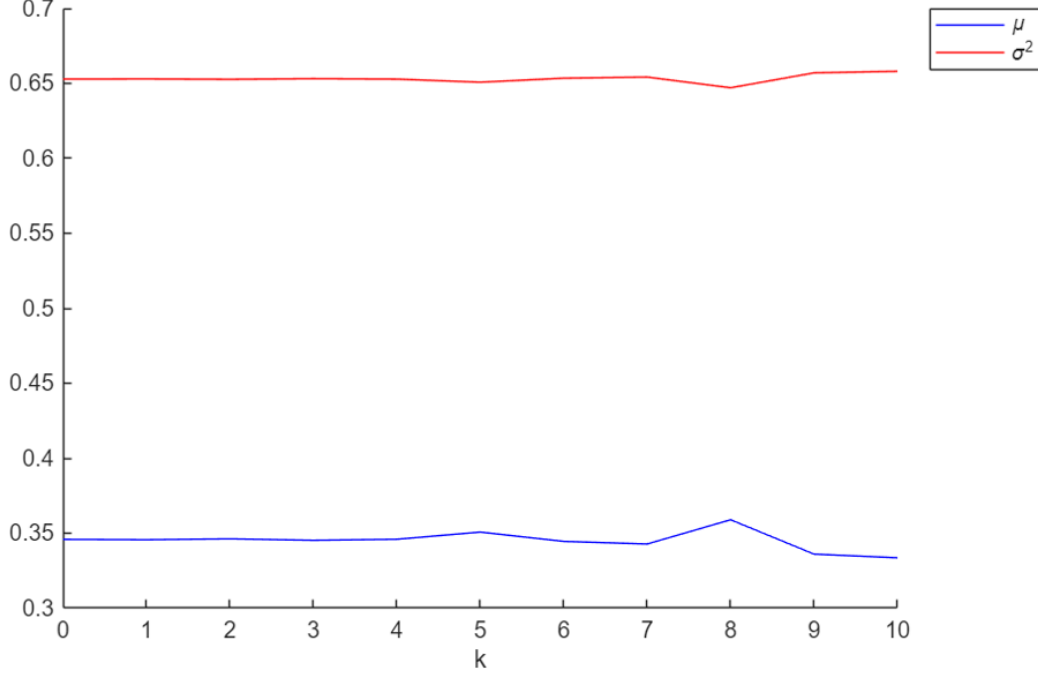


Figure 2: Graph of $\hat{\mu}_N^h$ (blue) and $(\hat{\sigma}^2)_N^h$ (red) against k

From Figure (2), there is no clear trend for the variation of $\hat{\mu}_N^h$ and $(\hat{\sigma}^2)_N^h$ with h ; both quantities appear to be approximately constant around $\hat{\mu} \approx 0.34$ and $\hat{\sigma}^2 \approx 0.65$, and any change in μ^h and $(\sigma^2)^h$ caused by the change in h is hidden by fluctuations caused by the variance of the estimators $\hat{\mu}_N^h$ and $(\hat{\sigma}^2)_N^h$.

We also observe that, for larger values of k (smaller values of h and N_k), there appears to be larger deviations of the estimator $\hat{\mu}_N^h$ from the value $\mu \approx 0.34$. This agrees with Equation (II) from Question (3); the variance $\mathbf{Var}[X_{Th-1}^h]$ remains approximately constant with k (since our estimator $(\hat{\sigma}^2)_N^h$ is approximately constant), but the number of samples N_k decreases as k increases, causing $\mathbf{Var}[\hat{\mu}_N^h]$ to increase with k .

As in Question (2), when h is sufficiently small, for $X_0^h \in [-1, x_2)$, we expect X_{Th-1}^h to approach x_1 as $Th^{-1} \rightarrow \infty$, and for $X_0^h \in (x_2, 1]$, we expect X_{Th-1}^h to approach x_3 as $Th^{-1} \rightarrow \infty$. For T sufficiently large, since X_0^h is uniformly distributed on $[-1, 1]$, we therefore expect

$$\begin{aligned}
\mu^h &\approx \frac{x_2+1}{2}x_1 + \frac{1-x_2}{2}x_3 \\
&= \frac{1}{2} \left[\left(1 + \cos\left(\frac{2\pi-\theta}{3}\right) \right) \cos\left(\frac{2\pi+\theta}{3}\right) + \left(1 - \cos\left(\frac{2\pi-\theta}{3}\right) \right) \cos\left(\frac{\theta}{3}\right) \right] \\
(\sigma^2)^h &\approx \frac{x_2+1}{2}x_1^2 + \frac{1-x_2}{2}x_3^2 - (\mu^h)^2 \\
&= \frac{1}{4} \left[1 - \cos^2\left(\frac{2\pi-\theta}{3}\right) \right] \left[\cos\left(\frac{\theta}{3}\right) - \cos\left(\frac{2\pi+\theta}{3}\right) \right]^2
\end{aligned} \tag{III}$$

Substituting in $\theta = \frac{\pi}{4}$ gives us

$$\begin{aligned}
\mu^h &\approx 0.345915873497... \\
(\sigma^2)^h &\approx 0.652884526419...
\end{aligned}$$

This value agrees with our results from Table (2).

Question 5.

Let $\mathbf{E}[T(X)] = \mu$, which is a constant.

$$\begin{aligned}
\mathbf{MSE}[T; \tau] &\triangleq \mathbf{E}[(T - \tau)^2] \\
&= \mathbf{E}[(T - \mu + \mu - \tau)^2] \\
&= \mathbf{E}[(T - \mu)^2] + 2\mathbf{E}[(T - \mu)(\mu - \tau)] + \mathbf{E}[(\mu - \tau)^2] \\
&= \mathbf{Var}[T] + 2(\mu - \tau)\mathbf{E}[T - \mu] + (\mu - \tau)^2 \quad \text{since } \mu, \tau \text{ do not depend on } X \\
&= (\mathbf{E}[T] - \tau)^2 + \mathbf{Var}[T] \quad \text{as required.}
\end{aligned}$$

Question 6.

From Equation (7), we have

$$\begin{aligned}
\mathbf{MSE}[\hat{\mu}_N^h] &= (\mathbf{E}[\hat{\mu}_N^h] - \mu)^2 + \mathbf{Var}[\hat{\mu}_N^h] \\
&= (\mathbf{E}[Y^i] - \mu)^2 + \frac{1}{N} \mathbf{Var}[Y^i] \quad \text{using the equations derived in Question (3)} \\
&= (\mathbf{E}[X_{T^{h-1}}^h] - \mu)^2 + \frac{1}{N} \mathbf{Var}[X_{T^{h-1}}^h] \quad \text{since } Y^i \text{ is distributed as } X_{T^{h-1}}^h \\
&= |\mu^h - \mu|^2 + \frac{1}{N} \mathbf{Var}[X_{T^{h-1}}^h] \\
&\leq A_1^2 h^2 + \frac{A_2}{N} \quad \text{as required.}
\end{aligned}$$

Suppose we use our full budget, i.e. we choose (N, h) such that $N \cdot \frac{A_3 T}{h} = C$. Then $N = \frac{Ch}{A_3 T}$. From the expression above,

$$\begin{aligned}
\mathbf{MSE}[\hat{\mu}_N^h] &\leq M(h) \triangleq A_1^2 h^2 + \frac{A_2}{N} \\
&= A_1^2 h^2 + \frac{A_2 A_3 T}{Ch}
\end{aligned}$$

To find the value of h which minimises this upper bound, we differentiate $M(h)$ and set it equal to 0:

$$\begin{aligned}
\frac{dM}{dh} &= 2A_1^2 h - \frac{A_2 A_3 T}{Ch^2} = 0 \\
\implies h^* &= \left(\frac{A_2 A_3 T}{2A_1^2 C} \right)^{1/3}
\end{aligned}$$

Then we have that the optimal MSE obeys

$$\begin{aligned}
\mathbf{MSE}[\hat{\mu}_N^h] &\leq M(h^*) = A_1^2 (h^*)^2 + \frac{A_2 A_3 T}{Ch^*} \\
&= 3 \left(\frac{A_1 A_2 A_3 T}{2C} \right)^{2/3}
\end{aligned}$$

and so the optimal MSE is $O(C^{-2/3}) \rightarrow 0$ as $C \rightarrow \infty$, as expected.

4 Multi-Level Monte Carlo

Question 7.

The series $\mu = \sum \mathbf{E}[Y_l]$ is absolutely convergent if the series $\sum |\mathbf{E}[Y_l]|$ converges. First, since $|\mathbf{E}[Y_l]| \geq 0$, the series of absolute values is monotone increasing, so it suffices to prove that it is bounded above. Observe that, for $l > 0$,

$$\begin{aligned} |\mathbf{E}[Y_l]| &= \left| \mathbf{E} \left[X_{Th_l^{-1}}^{h_l} \right] - \mathbf{E} \left[X_{Th_{l-1}^{-1}}^{h_{l-1}} \right] \right| \\ &= |\mu^{h_l} - \mu^{h_{l-1}}| \\ &= |(\mu^{h_l} - \mu) - (\mu^{h_{l-1}} - \mu)| \\ &\leq |\mu^{h_l} - \mu| + |\mu^{h_{l-1}} - \mu| \quad \text{by the triangle inequality} \\ &\leq A_1 h_l + A_1 h_{l-1} \end{aligned}$$

Then

$$\begin{aligned} \sum_{l=0}^N |\mathbf{E}[Y_l]| &= \mu^{h_0} + \sum_{l=1}^N |\mathbf{E}[Y_l]| \\ &\leq \mu^{h_0} + \sum_{l=1}^N A_1 (h_l + h_{l-1}) \\ &\leq \mu^{h_0} + 2A_1 \sum_{l=0}^{N-1} h_l \quad \text{since } h_l < h_{l-1} \\ &= \mu^{h_0} + 0.2A_1 \sum_{l=0}^{N-1} 2^{-l} \\ &\leq \mu^{h_0} + 0.4A_1 \quad \text{since } \sum_{l=0}^{\infty} 2^{-l} = 2 \end{aligned}$$

So the series $\sum |\mathbf{E}[Y_l]|$ is absolutely convergent, as required.

The truncation error by taking $\mu \approx \sum_{l=0}^L \mathbf{E}[Y_l]$ is bounded above by

$$\begin{aligned} \left| \mu - \sum_{l=0}^L \mathbf{E}[Y_l] \right| &= \left| \sum_{l=L+1}^{\infty} \mathbf{E}[Y_l] \right| \\ &\leq \sum_{l=L+1}^{\infty} |\mathbf{E}[Y_l]| \\ &\leq \sum_{l=L+1}^{\infty} A_1 (h_l + h_{l-1}) \\ &= A_1 \left(h_L + 2 \sum_{l=L+1}^{\infty} h_l \right) \\ &= 0.1 \times 2^{-L} A_1 \left(1 + 2 \sum_{l=1}^{\infty} 2^{-l} \right) \\ &= \frac{0.3A_1}{2^L} \end{aligned}$$

Question 8.

We compute $\hat{\mu}_{N_{1:L}}$ for $L = 10$, using the given level sizes (i) $N_l \equiv 5$ and (ii) $N_l = 2^{L-l}$. From Equation (III), we expect to have

$$\mu \approx m(\theta) \triangleq \frac{1}{2} \left[\left(1 + \cos \left(\frac{2\pi - \theta}{3} \right) \right) \cos \left(\frac{2\pi + \theta}{3} \right) + \left(1 - \cos \left(\frac{2\pi - \theta}{3} \right) \right) \cos \left(\frac{\theta}{3} \right) \right]$$

We can plot $m(\theta)$ (shown in green on Figure (3)) and see that it is approximated by the estimator with level sizes given by (ii), with small but noticeable variations.

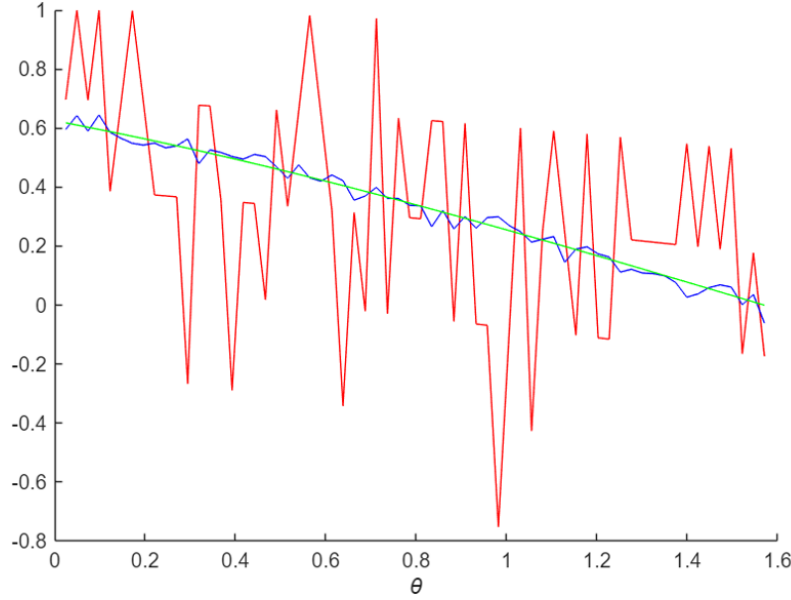


Figure 3: $\hat{\mu}_{N_{1:L}}$ against θ for (i) $N_l \equiv 5$ (red) and (ii) $N_l = 2^{L-l}$ (blue), with $m(\theta)$ (green) overlaid

From Figure (3), we observe that the estimator with level sizes $N_l = 5$ (red) appears to exhibit much greater variability than the estimator with level sizes $N_l = 2^{L-l}$ (blue). This is to be expected, as (i) involves $\sum_{l=0}^{10} 5 = 55$ samples Y_l^i , while (ii) involves $\sum_{l=0}^{10} 2^{10-l} = 2047$ such samples. We calculate the variance of

$\hat{\mu}_{N_{1:L}}$ for the two different estimators.

$$\begin{aligned}
\mathbf{Var} [\hat{\mu}_{N_{1:L}}] &= \mathbf{Var} \left[\sum_{l=0}^L \left(\frac{1}{N_l} \sum_{i=1}^{N_l} Y_l^i \right) \right] \\
&= \sum_{l=0}^L \frac{1}{N_l^2} \sum_{i=1}^{N_l} \mathbf{Var} [Y_l^i] \\
&= \sum_{l=0}^L \frac{1}{N_l} \mathbf{Var} [Y_l] \\
&= \frac{1}{N_0} \mathbf{Var} [X_{Th_0^{-1}}^{h_0}] + \sum_{l=1}^L \frac{1}{N_l} \mathbf{Var} [X_{2Th_{l-1}^{-1}}^{h_{l-1}/2} - X_{Th_{l-1}^{-1}}^{h_{l-1}}] \\
&\leq \frac{A_2}{N_0} + \sum_{l=1}^L \frac{A_4 h_{l-1}^2}{N_l} \\
&\leq \frac{A_2}{N_0} + \sum_{l=1}^{\infty} \frac{0.01 A_4}{2^{2l-2} N_l}
\end{aligned}$$

Substituting in the level sizes $\{N_l\}_{l=0}^L$ for (i) and (ii) and evaluating the sum gives us

$$\mathbf{Var} [\hat{\mu}_{N_{1:L}}] \leq \begin{cases} \frac{1}{5} A_2 + \frac{4}{1500} A_4 & \text{for (i)} \\ \frac{1}{1024} A_2 + \frac{1}{25600} A_4 & \text{for (ii)} \end{cases}$$

Thus the upper bound on the variance for (ii) is much smaller than the upper bound on the variance for (i) (by a factor of at least $\min\{\frac{1024}{5}, \frac{25600 \times 4}{1500}\} \approx 68$), agreeing with our observations about the variability of the estimators.

Question 9.

From Equation (7), we have

$$\mathbf{MSE} [\hat{\mu}_{N_{1:L}}] = (\mathbf{E} [\hat{\mu}_{N_{1:L}}] - \mu)^2 + \mathbf{Var} [\hat{\mu}_{N_{1:L}}]$$

Observe that

$$\mathbf{E} [\hat{\mu}_{N_{1:L}}] = \sum_{l=0}^L \left(\frac{1}{N_l} \sum_{i=1}^{N_l} \mathbf{E} [Y_l^i] \right) = \sum_{l=0}^L \mathbf{E} [Y_l]$$

Then, from Question (7), we have

$$\begin{aligned}
(\mathbf{E} [\hat{\mu}_{N_{1:L}}] - \mu)^2 &= \left| \sum_{l=L+1}^{\infty} \mathbf{E} [Y_l] \right|^2 \\
&\leq A_1^2 \left(h_L + 2 \sum_{l=L+1}^{\infty} h_l \right)^2 \\
&= \frac{0.09 A_1^2}{2^{2L}}
\end{aligned}$$

From Question (8), we have

$$\begin{aligned}
\mathbf{Var} [\hat{\mu}_{N_{1:L}}] &= \sum_{l=0}^L \frac{1}{N_l} \mathbf{Var}[Y_l] \\
&= \frac{1}{N_0} \mathbf{Var} [X_{Th_0^{-1}}^{h_0}] + \sum_{l=1}^L \frac{1}{N_l} \mathbf{Var} \left[X_{2Th_{l-1}^{-1}}^{h_{l-1}/2} - X_{Th_{l-1}^{-1}}^{h_{l-1}} \right] \\
&\leq \frac{A_2}{N_0} + \sum_{l=1}^L \frac{A_4 h_{l-1}^2}{N_l} \\
&= \frac{A_2}{N_0} + \sum_{l=1}^L \frac{0.04A_4}{2^{2l}N_l}
\end{aligned}$$

Then

$$\mathbf{MSE} [\hat{\mu}_{N_{1:L}}] \leq \frac{0.09A_1^2}{2^{2L}} + \frac{A_2}{N_0} + \sum_{l=1}^L \frac{0.04A_4}{2^{2l}N_l}$$

Question 10.

For $L = \infty$, we have

$$\mathbf{MSE} [\hat{\mu}_N] \leq \frac{A_2}{N_0} + \sum_{l=1}^{\infty} \frac{0.04A_4}{2^{2l}N_l}$$

The computational cost of running MLMC for given level sizes $\{N_l\}_{l=0}^{\infty}$ is

$$\begin{aligned}
\mathbf{Cost} (\hat{\mu}_N) &= \sum_{l=0}^{\infty} N_l \cdot \mathbf{Cost} (Y_l^i) \\
&= N_0 \cdot \mathbf{Cost} (X_{Th_0^{-1}}^{h_0}) + \sum_{l=1}^{\infty} N_l \cdot \left[\mathbf{Cost} (X_{Th_{l-1}^{-1}}^{h_l}) + \mathbf{Cost} (X_{Th_{l-1}^{-1}}^{h_{l-1}}) \right] \\
&= N_0 A_3 T h_0^{-1} + \sum_{l=1}^{\infty} N_l A_3 T [h_l^{-1} + h_{l-1}^{-1}] \\
&= 100A_3 \left[N_0 + \frac{3}{2} \sum_{l=1}^{\infty} 2^l \cdot N_l \right]
\end{aligned}$$

We assume that we use our full computational budget, so $\mathbf{Cost} (\hat{\mu}_N) = C$. Define

$$\begin{aligned}
N &= (N_1, N_2, N_3, \dots) \\
F(N) &= \frac{A_2}{N_0} + \sum_{l=1}^{\infty} \frac{0.04A_4}{2^{2l}N_l} \\
G(N) &= 100A_3 \left[N_0 + \frac{3}{2} \sum_{l=1}^{\infty} 2^l \cdot N_l \right] \\
H(N, \lambda) &= F(N) + \lambda(G(N) - C) \\
&= \frac{A_2}{N_0} + \sum_{l=1}^{\infty} \frac{0.04A_4}{2^{2l}N_l} + 100\lambda A_3 \left[N_0 + \frac{3}{2} \sum_{l=1}^{\infty} 2^l \cdot N_l \right] - \lambda C
\end{aligned}$$

We want to find the optimal $\tilde{N} = (\tilde{N}_1, \tilde{N}_2, \tilde{N}_3, \dots)$ that minimises $F(N)$ subject to $G(N) = C$, which amounts to finding stationary points $(\tilde{N}, \lambda)$ of $H(N, \lambda)$. Differentiating with respect to $\{N_l\}_{l=0}^{\infty}$ and λ and setting the derivatives equal to zero gives

$$\begin{aligned}
\frac{\partial H}{\partial N_0} &= -\frac{A_2}{N_0^2} + 100\lambda A_3 = 0 \\
\Rightarrow \tilde{N}_0 &= \sqrt{\frac{A_2}{100\tilde{\lambda}A_3}} \\
\frac{\partial H}{\partial N_l} &= -\frac{0.04A_4}{2^{2l}N_l^2} + 2^l \cdot 150\lambda A_3 = 0 \\
\Rightarrow \tilde{N}_l &= \sqrt{\frac{A_4}{3750 \cdot 2^{3l}\tilde{\lambda}A_3}} \quad \text{for } l \geq 1 \\
\frac{\partial H}{\partial \lambda} &= 100A_3 \left[N_0 + \frac{3}{2} \sum_{l=1}^{\infty} 2^l \cdot N_l \right] - C = 0 \\
\Rightarrow 100A_3 \left[\tilde{N}_0 + \frac{3}{2} \sum_{l=1}^{\infty} 2^l \cdot \tilde{N}_l \right] &= C
\end{aligned}$$

Let $\frac{1}{\sqrt{\tilde{\lambda}}} = \kappa$. Then

$$\begin{aligned}
100A_3 \left[\tilde{N}_0 + \frac{3}{2} \sum_{l=1}^{\infty} 2^l \cdot \tilde{N}_l \right] &= C \\
\Rightarrow 100A_3\kappa \left[\sqrt{\frac{A_2}{100A_3}} + \frac{3}{2} \sum_{l=1}^{\infty} 2^l \sqrt{\frac{A_4}{3750 \cdot 2^{3l}A_3}} \right] &= C \\
\Rightarrow 10\sqrt{A_3}\kappa \left[\sqrt{A_2} + \sqrt{\frac{3A_4}{50}} \sum_{l=1}^{\infty} 2^{-l/2} \right] &= C \\
\Rightarrow \kappa &= \frac{C}{10\sqrt{A_3}(\sqrt{A_2} + \frac{1}{\sqrt{2}-1}\sqrt{\frac{3A_4}{50}})}
\end{aligned}$$

Then the optimal level sizes are

$$\begin{aligned}
\tilde{N}_0 &= \frac{\kappa\sqrt{A_2}}{10\sqrt{A_3}} = \frac{C\sqrt{A_2}}{100A_3(\sqrt{A_2} + \frac{1}{\sqrt{2}-1}\sqrt{\frac{3A_4}{50}})} \quad \text{for } l = 0 \\
\tilde{N}_l &= \frac{\kappa\sqrt{A_4}}{\sqrt{3750 \cdot 2^{3l}A_3}} = \frac{C\sqrt{A_4}}{10\sqrt{3750 \cdot 2^{3l}A_3}(\sqrt{A_2} + \frac{1}{\sqrt{2}-1}\sqrt{\frac{3A_4}{50}})} \quad \text{for } l \geq 1
\end{aligned}$$

Question 11.

Observe that $\tilde{N}_l$ is monotonically decreasing with l for $l \geq 1$. Taking $N_l = \lfloor \tilde{N}_l \rfloor$, we see that

$$L = \max\{l \in \mathbb{N} : \tilde{N}_l \geq 1\}$$

Now

$$\begin{aligned}
N_l &\geq 1 \\
\Rightarrow \sqrt{2^{3l}} &\leq \frac{\kappa\sqrt{A_4}}{\sqrt{3750A_3}} \\
\Rightarrow l &\leq \frac{2}{3\log 2} \log \left[\frac{\kappa\sqrt{A_4}}{\sqrt{3750A_3}} \right]
\end{aligned}$$

Thus we have

$$L = \left\lfloor \frac{2}{3 \log 2} \log \left[\frac{\kappa \sqrt{A_4}}{\sqrt{3750 A_3}} \right] \right\rfloor = \left\lfloor \frac{2}{3 \log 2} \log \left[\frac{C \sqrt{A_4}}{10 \sqrt{3750 A_3} (\sqrt{A_2} + \frac{1}{\sqrt{2-1}} \sqrt{\frac{3A_4}{50}})} \right] \right\rfloor$$

and so $L = O(\log C)$.

Since L is now finite, we have

$$\begin{aligned} \mathbf{MSE} [\hat{\mu}_N] &= (\mathbf{E} [\hat{\mu}_N] - \mu)^2 + \mathbf{Var} [\hat{\mu}_N] \\ &\leq \frac{0.09 A_1^2}{2^{2L}} + \frac{A_2}{N_0} + \sum_{l=1}^L \frac{0.04 A_4}{2^{2l} N_l} \\ &\leq 0.36 A_1^2 \left[\frac{\sqrt{3750 A_3}}{\kappa \sqrt{A_4}} \right]^{\frac{4}{3}} + \frac{20 \sqrt{A_2 A_3}}{\kappa} + \frac{0.08 \sqrt{3750 A_3 A_4}}{\kappa} \sum_{l=1}^L \frac{1}{2^{l/2}} \quad (*) \end{aligned}$$

so $\mathbf{MSE} [\hat{\mu}_N] = O(\kappa^{-1}) = O(C^{-1}) \rightarrow 0$ as $C \rightarrow \infty$. Note that, in the step (*), we have made use of the fact that

$$\begin{aligned} N_l &= \lfloor \tilde{N}_l \rfloor \geq \frac{1}{2} \tilde{N}_l \quad \text{for } l \leq L \\ \text{and } L &= \left\lfloor \frac{2}{3 \log 2} \log \left[\frac{C \sqrt{A_4}}{10 \sqrt{3750 A_3} (\sqrt{A_2} + \frac{1}{\sqrt{2-1}} \sqrt{\frac{3A_4}{50}})} \right] \right\rfloor \\ &\geq \frac{2}{3 \log 2} \log \left[\frac{C \sqrt{A_4}}{10 \sqrt{3750 A_3} (\sqrt{A_2} + \frac{1}{\sqrt{2-1}} \sqrt{\frac{3A_4}{50}})} \right] - 1 \end{aligned}$$

We can see that the upper bound on the optimal MSE of the MLMC estimator decreases more rapidly as $C \rightarrow \infty$ compared to that of the MC estimator from Questions (4)-(6), which had optimal MSE bounded by $\mathbf{MSE} [\hat{\mu}_N^h] = O(C^{-2/3})$.

5 Application to Double-Well Loss Function

Question 12.

We have $\min\{|m_1(\theta) - X_T^h|, |m_2(\theta) - X_T^h|\} \approx 0$ for any initial point $x_0 \in [-1, 1]$. In other words, for any initial point $x_0 \in [-1, 1]$, h is sufficiently small that either $X_T^h \rightarrow m_1(\theta)$ or $X_T^h \rightarrow m_2(\theta)$ as $T \rightarrow \infty$, and T is sufficiently large for convergence to take place. Then

$$p_1(\theta) + p_2(\theta) = \mathbf{P} \left[\left\{ \lim_{T \rightarrow \infty} X_T^h = m_1(\theta) \right\} \cup \left\{ \lim_{T \rightarrow \infty} X_T^h = m_2(\theta) \right\} \right] = 1$$

Also, we have

$$\begin{aligned} \mu^h &= \mathbf{E}[X_T^h] \\ &= \mathbf{P} \left[\lim_{T \rightarrow \infty} X_T^h = m_1(\theta) \right] \mathbf{E} \left[X_T^h \mid \lim_{T \rightarrow \infty} X_T^h = m_1(\theta) \right] + \mathbf{P} \left[\lim_{T \rightarrow \infty} X_T^h = m_2(\theta) \right] \mathbf{E} \left[X_T^h \mid \lim_{T \rightarrow \infty} X_T^h = m_2(\theta) \right] \\ &\approx p_1(\theta) m_1(\theta) + p_2(\theta) m_2(\theta) \end{aligned}$$

where the approximation follows from the fact that

$$\begin{aligned} \min\{|m_1(\theta) - X_T^h|, |m_2(\theta) - X_T^h|\} &\approx 0 \\ \implies \mathbf{E} \left[X_T^h \mid \lim_{T \rightarrow \infty} X_T^h = m_i(\theta) \right] &\approx m_i(\theta) \end{aligned}$$

Thus we get

$$\begin{aligned} p_1(\theta) + p_2(\theta) &= 1 \\ m_1(\theta)p_1(\theta) + m_2(\theta)p_2(\theta) &= \mu \end{aligned}$$

which has solutions

$$\begin{aligned} p_1(\theta) &= \frac{m_2(\theta) - \mu}{m_2(\theta) - m_1(\theta)} \\ p_2(\theta) &= \frac{\mu - m_1(\theta)}{m_2(\theta) - m_1(\theta)} \end{aligned}$$

Question 13.

From Question (10), we have

$$\tilde{N}_l = \sqrt{\frac{2A_4}{75 \cdot 2^{3l}A_2}} \tilde{N}_0 \quad \text{for } l \geq 1$$

Recall that A_2, A_4 satisfy

$$\begin{aligned} \mathbf{Var}[X_{Th-1}^h] &\leq A_2 \\ \mathbf{Var}[X_{Th-1}^h - X_{2Th-1}^{h/2}] &\leq A_4 h^2 \end{aligned}$$

We know that X_{Th-1}^h takes values in $[-1, 1]$. Then $\mathbf{E}[(X_{Th-1}^h)^2], \mathbf{E}[X_{Th-1}^h]^2 \in [0, 1]$, so

$$\mathbf{Var}[X_{Th-1}^h] = \mathbf{E}[(X_{Th-1}^h)^2] - \mathbf{E}[X_{Th-1}^h]^2 \leq 1$$

Thus, we know $A_2 \leq 1$. (If θ is known, Equation (III) gives us an approximation for $\mathbf{Var}[X_{Th-1}^h]$).

Now $\mathbf{Var}[X_{Th-1}^h - X_{2Th-1}^{h/2}] \leq \mathbf{E}\left[\left(X_{Th-1}^h - X_{2Th-1}^{h/2}\right)^2\right]$. Since $f'_\theta(x) = 4x^3 - 3x - \cos(\theta)$, we have $f'_\theta(x) - f'_\theta(y) = (x - y)[4(x^2 + xy + y^2) - 3] = (x - y)g(x; y)$. We note that, for $x, y \in [-1, 1]$,

$$\begin{aligned} f'_\theta(x) &\in [-1 - \cos(\theta), 1 - \cos(\theta)] \subseteq [-2, 2] \\ 4(x^2 + xy + y^2) - 3 &\in [-3, 9] \end{aligned}$$

Then, for any $t \in \mathbb{N}$, we have

$$\begin{aligned} |X_t^h - X_{2t}^{h/2}| &= \left| (X_{t-1}^h - hf'_\theta(X_{t-1}^h)) - \left(X_{2t-2}^{h/2} - \frac{h}{2}hf'_\theta(X_{2t-1}^{h/2}) - \frac{h}{2}hf'_\theta(X_{2t-2}^{h/2}) \right) \right| \\ &\leq \left| X_{t-1}^h - X_{2t-2}^{h/2} - h \left(f'_\theta(X_{t-1}^h) - f'_\theta(X_{2t-2}^{h/2}) \right) \right| + h \left| \frac{1}{2}f'_\theta(X_{2t-1}^{h/2}) - \frac{1}{2}f'_\theta(X_{2t-2}^{h/2}) \right| \\ &\leq \left(1 - hg(X_{t-1}^h; X_{2t-2}^{h/2}) \right) |X_{t-1}^h - X_{2t-2}^{h/2}| + \frac{9h}{2} |X_{2t-1}^{h/2} - X_{2t-2}^{h/2}| \\ &\leq (1 + 3h) |X_{t-1}^h - X_{2t-2}^{h/2}| + \frac{9h}{2} |hf'_\theta(X_{2t-2}^{h/2})| \\ &\leq (1 + 3h) |X_{t-1}^h - X_{2t-2}^{h/2}| + 9h^2 \\ &\leq 9h^2 \sum_{k=0}^{t-1} (1 + 3h)^k \text{ by induction, since } X_0^h = X_0^{h/2} \\ &= 3h [(1 + 3h)^t - 1] \end{aligned}$$

Thus

$$\begin{aligned} \left(X_{Th^{-1}}^h - X_{2Th^{-1}}^{h/2} \right)^2 &\leq 9h^2 \left[(1+3h)^{Th^{-1}} - 1 \right]^2 \\ &\leq 9h^2 [e^{3T} - 1]^2 \end{aligned}$$

and we can take $A_4 \leq 9[e^{30} - 1]^2$.

We observe that this upper bound on the value of A_4 gives us $\lfloor \tilde{N}_0 \rfloor = 0$ for any reasonable number of total iterations $\sum_{i=0}^{\infty} \lfloor \tilde{N}_i \rfloor$. To avoid this, we instead take our optimal level sizes to be proportional to $\sqrt{\frac{1}{2^{3l}}}$ for all $l \geq 0$ (equal to our optimal level sizes as derived in Question (10), except for the N_0 term). Taking $N_0 = 32768$ gives us the following results for $\hat{\mu}$ and the corresponding computed values of $\hat{p}_1, \hat{p}_2$, calculated using our results from Question (12).

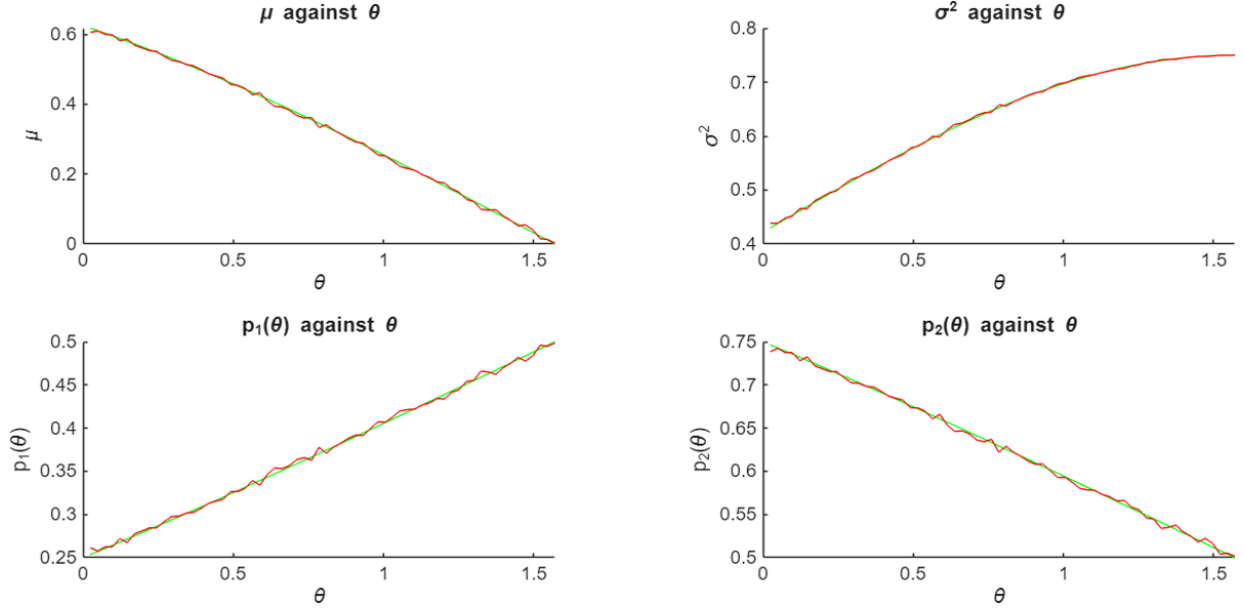


Figure 4: $\hat{\mu}, \hat{\sigma}^2, \hat{p}_1, \hat{p}_2$ against θ for $\theta \in \{\frac{k\pi}{2^7}\}_{k=1}^{2^6}$

Here, we have also used our derived values $\hat{p}_1, \hat{p}_2$ to compute the estimator $\hat{\sigma}^2$ of the variance $\mathbf{Var} [X_{Th^{-1}}^h]$, using the formula

$$\mathbf{Var} [X_{Th^{-1}}^h] = \mathbf{E} \left[(X_{Th^{-1}}^h)^2 \right] - (\mathbf{E} [X_{Th^{-1}}^h])^2 = \hat{p}_1 m_1^2 + \hat{p}_2 m_2^2 - \hat{\mu}^2$$

In Figure (4), the lines in red correspond to the computed values, while the lines in green correspond to the theoretical values, given by (as in Question (4))

$$\begin{aligned} p_1(\theta) &= \frac{x_2 + 1}{2} \\ p_2(\theta) &= \frac{1 - x_2}{2} \\ \mu(\theta) &= p_1 x_1 + p_2 x_3 \\ \sigma^2(\theta) &= p_1 x_1^2 + p_2 x_3^2 - \mu^2 \\ &= p_1 p_2 (x_1 - x_3)^2 \quad \text{since } p_1 + p_2 = 1 \end{aligned}$$

From this, we can see that the greatest variance in the outcome of gradient descent (as measured by $\hat{\sigma}^2$, the estimator of $\mathbf{Var}[X_{Th-1}^h]$) for $\theta \in [0, \frac{\pi}{2}]$ appears to occur at $\theta = \frac{\pi}{2}$.

From Equation (III),

$$\mathbf{Var}[X_{Th-1}^h] = \frac{1}{4} \sin^2\left(\frac{2\pi - \theta}{3}\right) \left[\cos\left(\frac{\theta}{3}\right) - \cos\left(\frac{2\pi + \theta}{3}\right) \right]^2$$

Then

$$\frac{d}{d\theta} (\mathbf{Var}[X_{Th-1}^h]) = \frac{1}{6} \sin\left(\frac{2\pi - \theta}{3}\right) \left[\cos\left(\frac{\theta}{3}\right) - \cos\left(\frac{2\pi + \theta}{3}\right) \right] \left[\cos\left(\frac{2\theta}{3}\right) - \cos\left(\frac{2\pi - 2\theta}{3}\right) \right]$$

which is positive for $\theta \in [0, \frac{\pi}{2})$ and has a zero at $\theta = \frac{\pi}{2}$, agreeing with our results above.

6 Program Listings

Gradient Descent Algorithm

```
%Defining parameters, locations of minima
theta = pi/6;
h = 0.01;
N = 1000;
x1 = cos(13*pi/18);
x3 = cos(pi/18);

for x_0 = -1:1/50:1 %Evenly spaced points initial points x_0
    x = x_0;

    for i = 1:N %Apply gradient descent starting from the initial points
        x = x - h.*df(x,theta);
    end

    d = min([abs(x-x1) abs(x-x3)]); %Distance from nearest minimum after N steps
    disp(['$' num2str(x_0) '$ & $' num2str(x,10) '$ & $' num2str(d) '$ \\''])

end
```

Monte Carlo Estimator of μ^h

```
theta = pi/4; %Parameters
resultm = zeros(11,1); %Initialising empty arrays
resultn = zeros(11,1);
resulth = zeros(11,1);

for k = 0:10
    h = 0.1 .* (2).^(-k); %h
    Nk = 2 .* (20-k); %N_k
    Thinv = 100 .* (2).^k; %Th^{-1}
    mean = 0;
    meansq = 0;

    for i = 1:Nk
        x = 2.*rand() - 1; %Randomly generate an initial point x_0
        for j = 1:Thinv %Apply gradient descent to x_0
            x = x - h.*df(x,theta);
        end
        mean = mean + x./Nk; %Update the mean value of X after Th^{-1} iterations
        meansq = meansq + (x.^2)./Nk; %Update the mean value of X^2 after Th^{-1} iterations
    end

    resultm(k+1,1) = mean; %Stores E[X^{h}_{Th^{-1}}]
    resultn(k+1,1) = meansq - mean.^2; %Stores Var[X^{h}_{Th^{-1}}]
    resulth(k+1,1) = h; %Stores values of h

    disp(['$' num2str(k) '$ & $' num2str(h) '$ & $' num2str(resultm(k+1,1)) ...
        '$ & $' num2str(resultn(k+1,1)) '$ \\''])

end
```

Multi-level Monte Carlo Estimator of μ

```

L=10; %Parameter
resulti = zeros(2^6,1); %Initialising empty arrays
resultii = zeros(2^6,1);
thetacoord = zeros(2^6,1);

for k = 1:2^6
    theta = k.* pi ./ (2.^7); %Given values of theta
    thetacoord(k,1) = theta;

    for l = 0:L
        Nli = 5; %N_l under (i)
        Nlil = 2.^ (L-1); %N_l under (ii)

        h = 0.1 .* (2).^(-1);
        t = 100 .* (2).^1; %Th^{-1}

        for i = 1:Nli
            x = 2.*rand() - 1; %Generate X_0
            y = fgd(x,theta,h,t); %Apply gradient descent to get X^{h}_{Th^{-1}}
            z = 0;
            if l ~= 0 %Exclude l=0 case
                z = fgd(x,theta, 2*h, t/2); %Compute X^{h/2}_{2Th^{-1}}
            end
            resulti(k,1) = resulti(k,1) + (y-z) ./ Nli; %Contribution to mean
        end

        for i = 1:Nlil
            x = 2.*rand() - 1; %Generate X_0
            y = fgd(x,theta,h,t); %Apply gradient descent to get X^{h}_{Th^{-1}}
            z = 0;
            if l ~= 0 %Exclude l=0 case
                z = fgd(x,theta, 2*h, t/2); %Compute X^{h/2}_{2Th^{-1}}
            end
            resultii(k,1) = resultii(k,1) + (y-z) ./ Nlil; %Contribution to mean
        end
    end
end

```

Multi-level Monte Carlo Estimator of p_1, p_2

```

L=10; %Greatest value of l for which N_l > 1
result = zeros(2^6,1); %Initialise empty arrays
resultsq = zeros(2^6,1);
thetacoord = zeros(2^6,1);

Nl = zeros(L+1,1);
Nl = [32768 11585 4096 1448 512 181 64 22 8 2 1]; %Values for optimal N_l
%Proportional to 2^{-3l/2}

for k = 1:2^6
    theta = k.* pi ./ (2.^7);
    thetacoord(k,1) = theta;

    for l = 0:L %MLMC scheme from before
        h = 0.1 .* (2).^(-l);
        t = 100 .* (2).^l;
        for i = 1:Nl(l+1)
            x = 2.*rand() - 1;
            y = fgd(x,theta,h,t);
            z = 0;
            if l ~= 0
                z = fgd(x,theta, 2*h, t/2);
            end
            result(k,1) = result(k,1) + (y-z) ./ Nl(l+1);
        end
    end
end

x1 = cos((2*pi + thetacoord)./3);
x2 = cos((2*pi - thetacoord)./3);
x3 = cos(thetacoord ./ 3);

p1 = (x3 - result) ./ (x3 - x1); %Computed value of p_1 from mu
p2 = (result - x1) ./ (x3 - x1); %Computed value of p_2 from mu

```